Fluids and Lubricants Specifications

MTU Fluids and Lubricants Specifications for Series 1800

A001062/02E

Power. Passion. Partnership.

Printed in Germany

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

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DCL-ID: 0000014456 - 001

1 Preliminary Remarks

1.1 General information

The service life, operational reliability and function of the drive systems are largely dependent on the fluids and lubricants employed. The correct selection and treatment of these fluids and lubricants are therefore extremely important. This publication specifies which fluids and lubricants are to be used.

The Fluids and Lubricants Specifications will be amended or supplemented as necessary. Before using them, make sure you have the latest version. The latest version is also available at: http://www.mtu-online.com/mtu/Select language: English/MTU_ValueCare/MTU_ValueService Technical Documentation/ Fluids and Lubricants Specifications/Fluids and Lubricants Specification - Valid for Series 1800 Powerpack.

If you have further queries, please contact your MTU representative.

The Fluids and Lubricants Specifications are applicable to PowerPacks with Series 1800 engines compliant with emission specifications
- Euro 3
- EU stage 3A / EPA Tier 3 (with diesel particulate filter but without SCR aftertreatment systems)
- EU Stage 3B

Test standards for fluids and lubricants:

DIN	Federal German Standards Institute
EN	European Standards
ISO	International Standards Organization
ASTM	American Society for Testing and Materials
IP	Institute of Petroleum

Note:

Use of the approved fluids and lubricants, either under the brand name or in accordance with the specifications given in this publication, constitutes part of the warranty conditions.

The supplier of the fluids and lubricants is responsible for the worldwide standard quality of the named products.

i	Fluids and lubricants for drive systems may be hazardous materials. Certain regulations must be obeyed when handling, storing and disposing of these substances.

These regulations are contained in the manufacturers' instructions, legal requirements and technical guidelines valid in the individual countries. Great differences can apply from country to country and a generally valid guide to applicable regulations for fluids and lubricants is therefore not possible within this publication.

Users of the products named in these specifications are therefore obliged to inform themselves of the locally valid regulations. MTU Friedrichshafen GmbH accepts no responsibility whatsoever for improper or illegal use of the fluids and lubricants which it has approved.

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2 Lubricants for Four-Cycle Engines

2.1 Lubricants

Engine oils

ℹ	Dispose of used fluids and lubricants in accordance with local regulations.

Selection of viscosity grades

Selection of the viscosity grade is based primarily on the ambient temperature at which the engine is to be started and operated. If the relevant performance criteria are observed the engines can be operated both with single grade and multigrade oils, depending on the application. Standard values for the temperature limits in each viscosity grade are shown in Chart 1.

If the prevailing temperature is too low, the engine oil must be preheated.

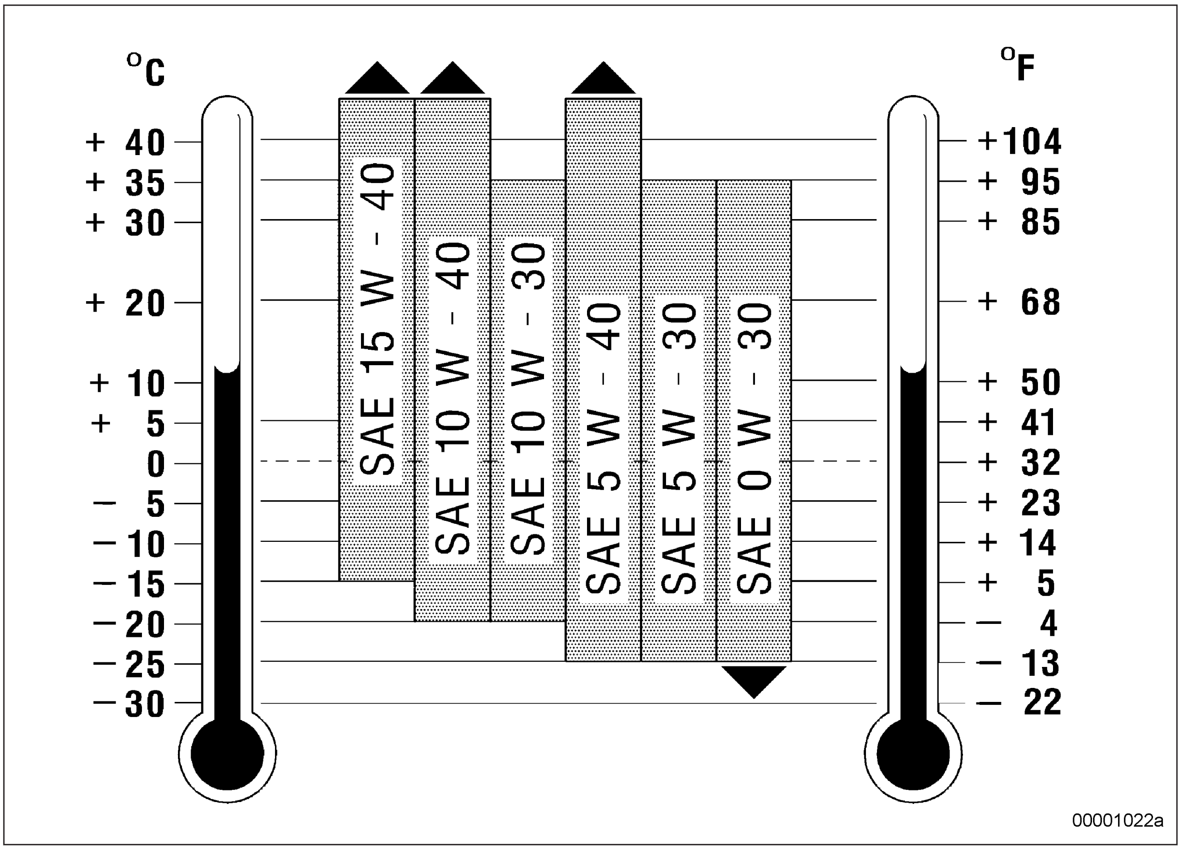

Chart 1

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Engine oils for Series 6H 1800 engines

For engine model 6H 1800, only engine oils in accordance with Chapter 5.2 (→ Page 20) may be used.

For engines equipped with diesel particulate filter, only "Low SPAsh oils" in accordance with Chapter 5.3 (→ Page 21) may be used!

The oil change interval is 1000 operating hours or max. 1 year under the condition that approved fuels as per Chapter 4 (→ Page 14) are used.

If fuels which have not been approved are used, shorter oil change intervals are to be expected.

Before using fuels which have not been approved, consult MTU Friedrichshafen GmbH to agree the oil change intervals.

Operating the engines with biodiesel/FAME acc. to EN14214:2010-04 involves reduced oil change intervals, see Chapter 4 (→ Page 14).

Mixing engine oils

Mixing different engine oils is strictly prohibited!

Topping up engine oil with a different oil brand than currently used in the engine is also strictly prohibited!

Changing to another oil grade can be done together with an oil change. The remaining oil quantity in the engine oil system is not critical in this regard.

Transmission oils for rail vehicles with ZF transmission

(Abstract from the ZF List of Lubricants TE-ML16, Edition 01.10.2008)

The ZF Lists of Lubricants are updated every three months on 01.01., 01.04., 01.07. and 01.10.. Before using them, make sure you have the latest version. The latest version is also available at:

http://www.zf.com/Corporate/en menu item Products & Services / Service Portfolio / Service Center / Lubricants & Steel / List of Lubricants / Please choose your Language / TE-ML 16.

Product groups automatic transmissions for rail vehicles	Lubricant classes for service fills[1] transmissions without/with ZF-Intarder
ASRail • 12 AS 2303. 12 AS 2703. 12 AS 3103. 16 AS2 603	16K / 16P
Ecomat • HP 500 R, HP 590 R, HP 600 R • HP 502 R, HP 592 R, HP 602 R	16L / 16M / 16N Automatic Transmission Fluid (ATF)[2]
Ecomat • HP902 R	16N
EcoLife (up to 105 °C)	16Q

[1] = Approved commercial products (see Chapter 5), oil change intervals and low temperature limits (specified below).

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[2] = Particularly recommended: The fully synthetic ATF ZF-Ecofluid A PLUS was developed specifically for use in Ecomat transmissions. This combination of a synthetic base oil with a specially balanced additive package delivers superlative oxidation stability and consistent friction characteristics. The viscosity level is ideally suited to this transmission and this factor, combined with resistance to scuffing and pitting, assures the unit of extremely valuable protection and a correspondingly long service life for its bearings and gears. Another positive feature of ZF-Ecofluid A PLUS is its flat viscosity characteristics curve which makes it particularly well suited to operation in cold as well as in hot climatic zones .

At greasing points pay attention to the manual!

ℹ	Additives of any kind added later to the oil change the oil in a manner that is unpredictable, and they are therefore not permitted. No liability of any kind will be accepted by ZF for any damage resulting from the use of such additives.

Oil change intervals for ASRail transmissions:

Lubricant classes[1]	Oil change interval [km / years] [2,3]
16K	300,000 km / every 2 years
16P	360,000 km / every 3 years

[1] = Pay attention to approved trade products and lubricant classes.

[2] = Oil change required, depending on what occurs first.

[3] = After consultation with the Customer Service department of ZF Friedrichshafen AG, Special Drive technology, and after an oil analysis has been made (after agreed mileages), longer oil change intervals can be applied to some reference transmissions. The procedure for taking oil samples is described in the respective Service Information.

Oil and filter change intervals for Ecomat transmissions HP 500 R, HP 590 R, HP 600 R, HP 502 R, HP592 R, HP 602 R for rail vehicles:

Lubricant classes[1]	Oil and filter change interval [km / years] [2,3]
16L	60,000 km / every 2 years
16M	120,000 km / every 2 years
16N	150,000 km / every 3 years

[1] = Pay attention to approved trade products and lubricant classes.

[2] = Oil change required, depending on what occurs first.

[3] = After consultation with the Customer Service department of ZF Friedrichshafen AG, Special Drive technology, and after an oil analysis has been made (after agreed mileages), longer oil change intervals can be applied to some reference transmissions. The procedure for taking oil samples is described in the respective Service Information.

Oil and filter change intervals for Ecomat transmissions HP 902 R for rail vehicles:

Lubricant classes[1]	Oil and filter change interval [km / years] [2,3]
16N	120,000 km / every 3 years

[1] = Pay attention to approved trade products and lubricant classes.

[2] = Oil change required, depending on what occurs first.

[3] = After consultation with the Customer Service department of ZF Friedrichshafen AG, Special Drive technology, and after an oil analysis has been made (after agreed mileages), longer oil change intervals can be applied to some reference transmissions. The procedure for taking oil samples is described in the respective Service Information.

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Oil and filter change intervals for EcoLife transmissions for rail vehicles:

Lubricant classes[1]	Oil and filter change interval [km / years] [2,3]
16Q	180,000 km / every 3 years

[1] = Pay attention to approved trade products and lubricant classes.

[2] = Oil change required, depending on what occurs first.

[3] = After consultation with the Customer Service department of ZF Friedrichshafen AG, Special Drive technology, and after an oil analysis has been made (after agreed mileages), longer oil change intervals can be applied to some reference transmissions. The procedure for taking oil samples is described in the respective Service Information.

The above oil change intervals only apply to complete fills. If oil is changed to another class of lubricants, the following oil and filter change intervals apply :

Changing class of lubricant from	Oil and filter change interval [km / years] [1]
16L => 16M	90,000 km / every 2 years
16L => 16N	120,000 km / every 2 years
16M => 16N	150,000 km / every 3 years

[1] = Oil change required, depending on what occurs first.

Application areas of lubricants

The following illustration shows application areas of the various SAE classes in relation to the ambient temperatures to be expected.

The oils have a bottom limit of max. dynamic viscosity (Brookfield) of 150,000 mPas, which roughly corresponds to the viscosity limit at low temperatures.

The upper limit is determined by the load in the transmission and the appearing temperature level during operation. It can be assumed that high ambient temperatures will also result in higher oil sump temperatures. For detailed information on the low temperature behavior of the specific product see at the safety data sheet of the supplier.

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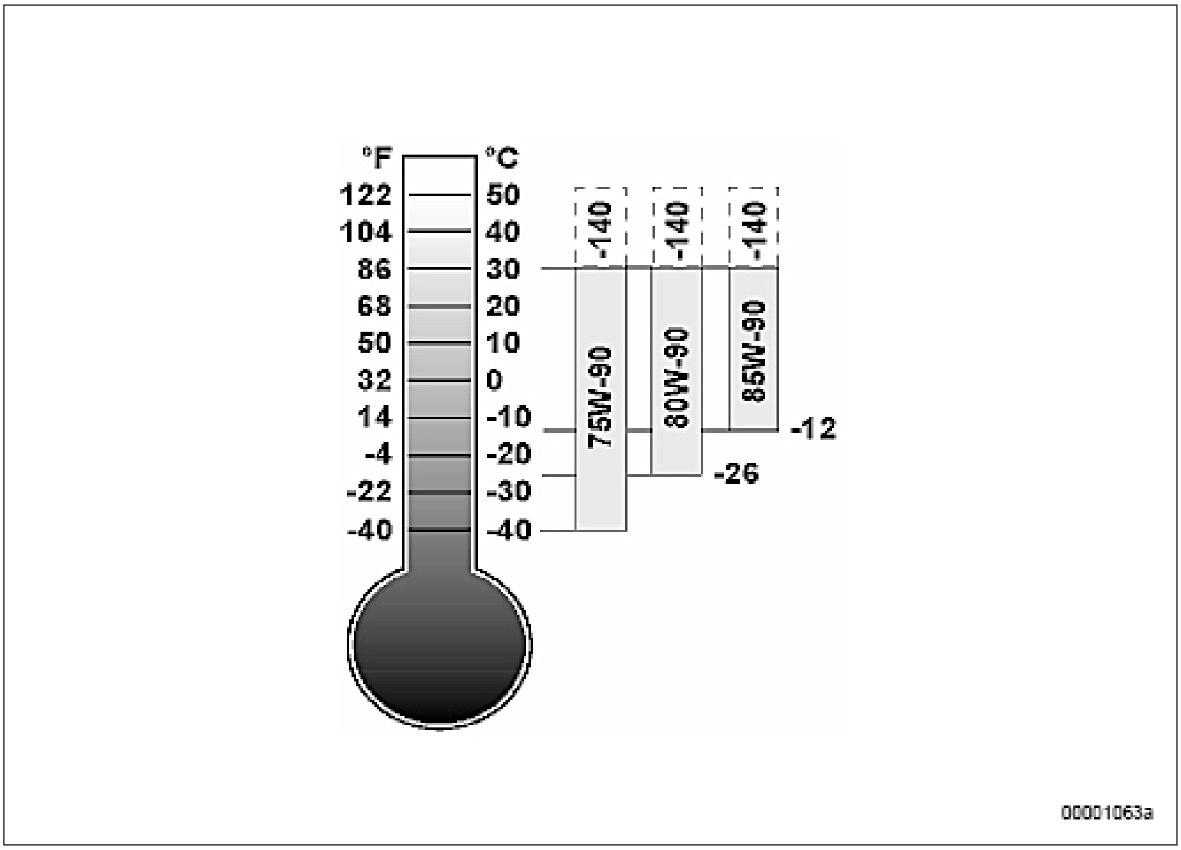

The user has to observe of the low temperature limits!

Lubricant classes	Viscosity grades	Use at oil sump temperature as over
16K / 16L / 16M / 16N / 16P / 16Q	75W-80 / 75W-85 / 75W-90 / 75W-110 / 75W-140 / ATF	- 40 °C

Power transmission oils for Voith turbo transmissions T 211 re.4 + KB190

Abstract from Voith documentation "Transmission oils for Voith turbo transmissions 120.00059000 Version 1 and Repair instructions 120.00068341 Version 3

Voith publications are continuously updated. Before using them, make sure you have the latest version. The latest version is also available at:

“www.Voithturbo.com” menu item Products & Applications / Rail / Hydrodynamic drives / Publications / Data sheet (Title) - (Market Division / Product Group) Hydrodynamic Drives / Transmission oils for turbo transmissions

Oil and filter change intervals for Voith turbo transmissions T 211 re4 + KB190

Oil and filter change interval based on running hours[1]	Oil and filter change intervals based on mileage (km)
5,000	300,000

[1] = Running hours are accumulated operating hours at speeds of more than 1 km/h.

Use at low temperatures

The approved transmission oils allow cold starts at temperatures down to -20 °C.

Special measures must be taken if temperatures are lower.

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Oil filtration

Ensure oil filtration to purity class 15/11 as per ISO 4406 when filling the transmission with oil. For appropriate filter units please contact Voith Turbo.

The maximum quantity of foreign particles in 100 ml oil for this purity class is:
- Particles >5µm: 32,000 (class 15)
- Particles >15µm: 2,000 (class 11)

Oil and filter replacement intervals for Voith transmission unit DIWA 884.5 / SWG

Oil and filter change intervals based on diesel engine operating hours	Oil and filter change intervals based on mileage (km)
4,000	120,000

Hydraulic system

ℹ	The oil change interval for the hydraulic system is 4000 operating hours / max. 2 years!

The approved engine oils specified in Chapter 5 must be used.

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